Project 1 Written Report 2

ME 370

10/9/26

Group S34

Will Garber, Kavish Mistry, Daksh Bhaskar, Kashvi Kunder

1.1 PATH SYNTHESIS

In Lab 5, we chose a double straight curve from the coupler-curve atlas with L1/L2 = 3.5, γ = 180°, and L3/L2 = L4/L2 = BP/L2 = 2.5. In the simulation, the crank is 10 mm, the ground link 35 mm, and the coupler and rocker 25 mm each, with P located 25 mm past B along line AB. This is a Grashof crank-rocker (10 + 35 ≤ 25 + 25), so the crank can rotate fully. Because γ = 180°, the leg is a single straight link. P traces a teardrop loop about 27 mm wide and 24 mm tall: its flatter lower edge is the stance stroke that pushes the rack, and its rounded upper lobe lifts the foot clear on the return.

1.1.1 Figures

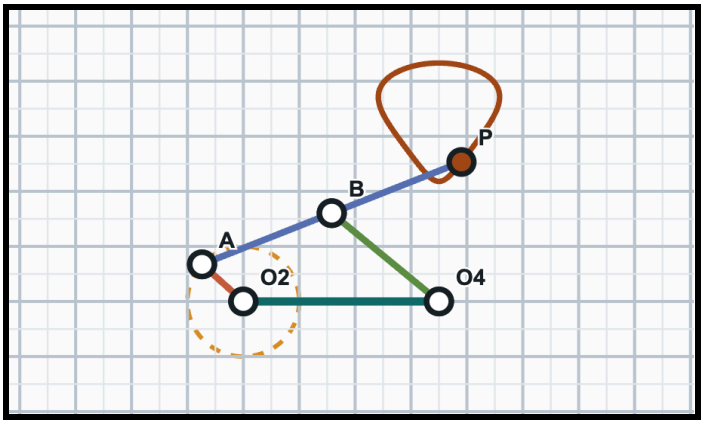


Figure 1: Simulated mechanism and ideal coupler path of P in the Fusion Mechanism Generator

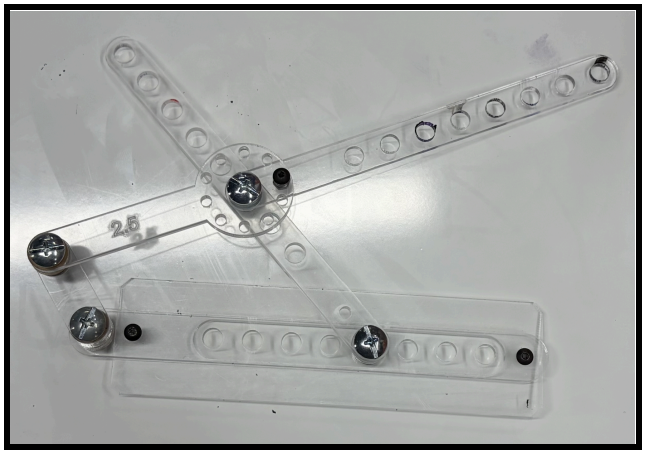


Figure 2: Lab 5 physical prototype, an adjustable acrylic linkage set to the 2.5 link ratio.

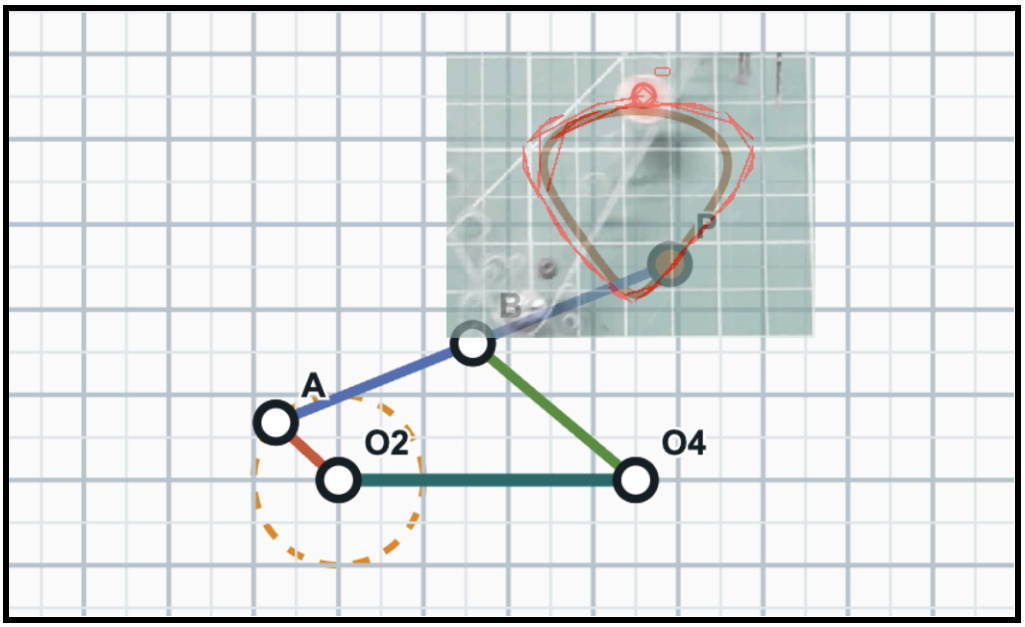


Figure 3:Tracked path of the physical prototype (red, from Tracker) overlaid on the simulated mechanism and path (transparent brown)

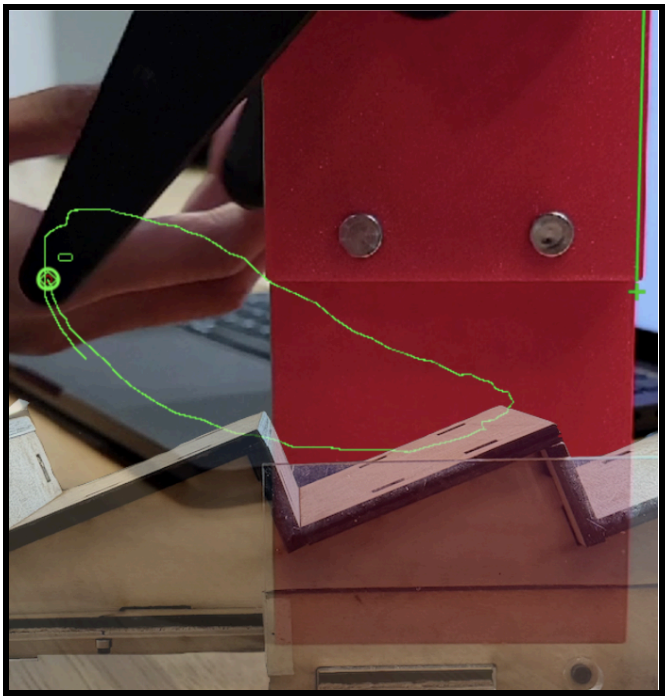


Figure 4: Tracked path of the physical prototype pushing the stair rack testing device (green).

1.1.2 Challenges during design

Finding a curve that had both a flat stance stroke long enough to push the rack and enough lift to clear the stair came with challenges during the design process. First, the acrylic kit only allowed link lengths at fixed hole positions, so the prototype approximates the atlas ratios rather than matching them exactly. Second, the bolts sit in clearance holes, which let the links shift and tilt out of plane during motion.

1.1.3 Simulated vs Physical Path

As Figure 2 shows, the tracked path keeps the simulated teardrop shape and tip location but deviates most around the upper lobe. The largest source of error was joint play due to excessive clearance between the links. The fixed hole spacing also forced approximate link lengths, which links cut to exact lengths would correct. The measurement setup added further error through camera angle and marker placement. Filming square to the plane of motion and clamping the ground link to the table would minimize these errors.

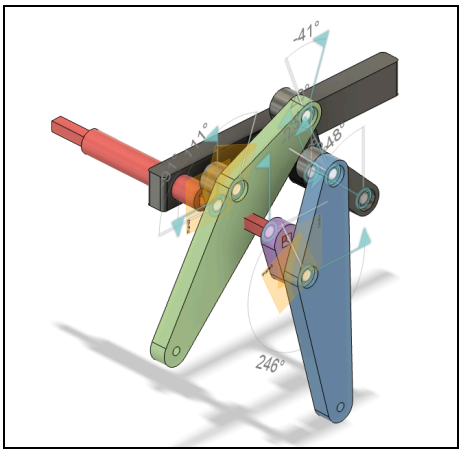
1.1.4 Fit to the Project 1

The path suits the task of pushing the spring-loaded stair rack forward with at least 2 N. The return path clears the feet of the stairs to prevent jamming. Running two legs 180° apart keeps one foot on the rack at all times, so the spring cannot push the rack back between strokes.

2.1 BACKGROUND & RESEARCH

2.1.1 Background

In Phase I, we selected a crank-rocker four-bar mechanism (L1 = 46 mm, L2 = 17 mm, L3 = 44 mm, L4 = 36 mm) to advance a spring-loaded stepped rack within a 24 cm cubic envelope and deliver at least 2 N of horizontal force. While a 3D-printed single-leg prototype confirmed full crank rotation and the intended foot path, it revealed link/joint fragility and ground-link deflection under load. Phase II goals include adding a second leg 180° out of phase, building a rigid frame, and strengthening the links/joints.



2.1.2 Two-Leg Interference and Frame Stability

The CAD model shown on the right is our first attempt at the two-legged design. Unfortunately, the orange crankshaft interfering with the green coupler (leg) during rotation is unavoidable with this one-sided frame design. This design is also structurally weak, since the overhung shaft and pivots deflect and twist under the load.

2.1.3 Rack Geometry Re-evaluation

While modeling the fixture in CAD, we realized we had referenced the stairs incorrectly. The stairs were positioned too high, which reduced the vertical rise and misaligned our horizontal run. Lowering the stairs to their proper position increased the required vertical rise and revealed the true required rise to be 88.1 mm. To match this larger rise and prevent the foot from landing shallower on each tooth, we uniformly rescaled all linkage lengths.

2.1.4 Joint and Printing Tolerances

The fragile pins and out-of-plane motion from Phase I led us to research how 3D-printed parts fit together. FDM holes print slightly undersize and pegs slightly oversize, so rotating joints need about 0.4 mm of diametral clearance. Printed square sockets have rounded internal corners from the circular nozzle, so square shafts need chamfered corners to seat. After comparing multiple fastener types, we found that barrel fasteners give a smooth, accurately sized bearing surface and remove the flexing printed pin.

2.2 IDEATION PROCESS & TOP DESIGN

2.2.1 Ideation Process

We first chose to place the two legs, 180° out of phase, between walls on both sides of the rack. A 3D simulation with a single link joining the cranks in the middle showed this removes the one-sided design's interference, since no shaft crosses a leg's path. We then replaced that link with two crank links joined by a short shaft across a gap, supported by a centre overhang from the top of the frame with a hole at its base for the shaft to rotate in. The crankshaft is therefore supported at both walls and the middle, so it does not bend under load.

To keep the legs close together while the frame still cleared the fixture's side walls, each frame wall was split into two connected parts: an outer part on the table outside the fixture, and an inner part raised above the fixture walls. Since the inner part clears the fixture, the linkage can sit closer together than the fixture's width would otherwise allow.

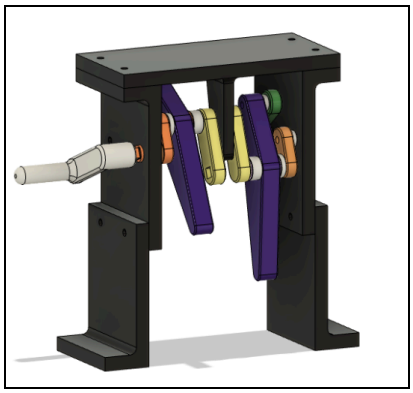
For the joints, we replaced the Phase I snap pins with stainless barrel fasteners, giving a smooth metal bearing surface that does not flex, and added spacers between links to remove axial play. The crank links have their connecting rod extruded as part of the link, reducing part count and removing a fit that would need careful tolerancing. Finally, we adjusted the link lengths so the stride matched the step pitch, keeping the trajectory's shape while improving how far the rack advances each turn.

2.2.2 Top Design

Table 1 : Design Comparison

Version	*Layout*	*Outcome*
1	*One-sided frame, crankshaft cantilevered through both legs*	*Rejected: crankshaft collides with the coupler leg; shaft bends under load*
2	*Two-sided frame, single component transferring torque between two legs*	*Rejected: Interference removed, but crankshaft supported only at the walls*
3	*Two-sided frame, paired crank links with a centrally supported shaft, stepped frame walls*	*Selected*

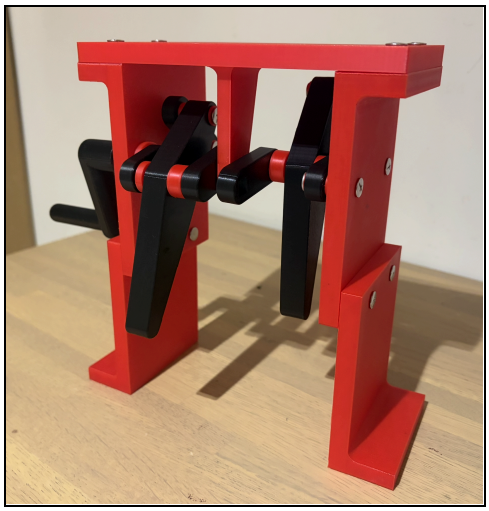
Version 3 on the right was selected as our final design. It keeps the interference-free layout of Version 2 while supporting the crankshaft at the centre as well as the walls, and its stepped frame walls bring the legs closer together without contacting the fixture.



2.3 FINAL PROTOTYPE & FUTURE PLANS

2.3.1 Final Prototype

The final prototype, shown on the right, is the physical build of Version 3, printed in PLA with a red frame and spacers and black links. The links were scaled to L1 = 50.7 mm, L2 = 18.7 mm, L3 = 48.5 mm and L4 = 39.6 mm to match the 88.1 mm step pitch, with the O2 and O4 pivots 153 mm above the table. The 15 mm thick legs resist out-of-plane bending, and the joints use barrel fasteners with spacers between links. A handle on the left wall drives both legs, and the frame's foot flanges clamp to the table on either side of the rack without contacting the fixture. On the spring-loaded rack, the prototype advanced the rack continuously as the crank turned, with the out-of-phase legs keeping constant contact
with the stairs.



2.3.2 Future Plan (Project 2)

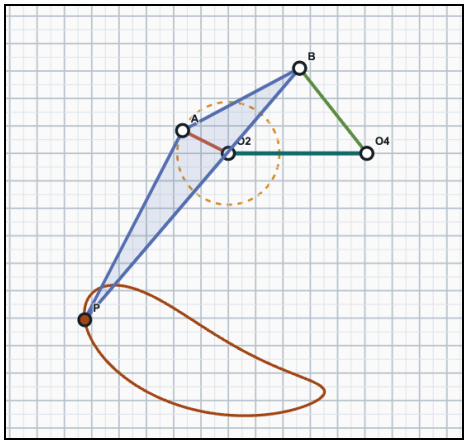
For Project TerraTrek 2, the mechanism must become a self-contained, motorized robot that climbs the steps itself while performing a synchronised secondary agricultural task. Our two-legged linkage carries forward as the basis for the robot's legs, and we plan to add further pairs of legs at different phases so there are at least 3 legs contacting the ground at any given time. Our current foot trajectory worked well for Project 1 but may need to be updated for project 3. The hand crank will be replaced by a battery-powered motor, adding complexity to the design. We will also have to consider what crankshaft-driven secondary task we will add.

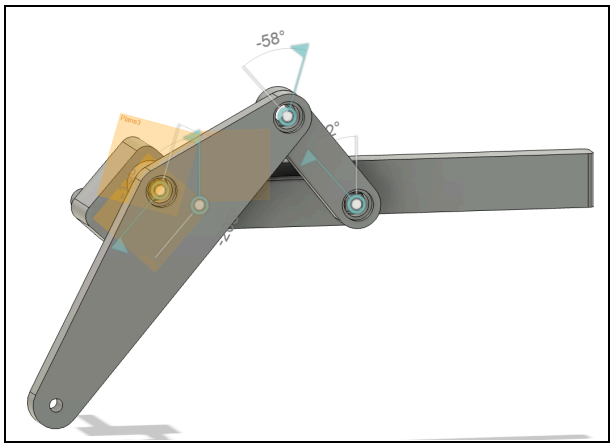
4. APPENDIX

Table 2. Expense Report

Description	Qty/$	Unit Price	Qty Purchased	Source	Purpose	Total Cost
PLA Filament	1g/$0.03	$0.03	271.16 g	Innovation Studio	9 links & Prototype	$8.13
PLA Filament (Personal Printer)	1g/$0.03	$0.03	230g	Personal	Walls	$6.90
Manufacturing Cost for Printing (Innovation Studio)	1cm^3/$0.025	$0.025	708 cm^3	Innovation Studio	9 links	$17.7
Manufacturing Cost for Printing (Personal)	1cm^3/$0.025	$0.025	578 cm^3	Personal	Walls	$14.45
Barrel Fasteners	1 set (30 fasteners) /$4	$4	1 set	Personal Purchase	Between links and for wall	$4.00
Personal Cost : $25.35 | Innovation Studio Cost : $25.83						$51.18

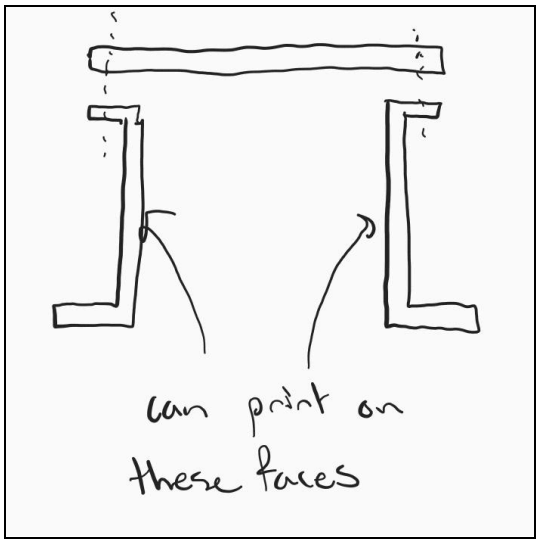
4.2 ORIGINAL TRAJECTORY

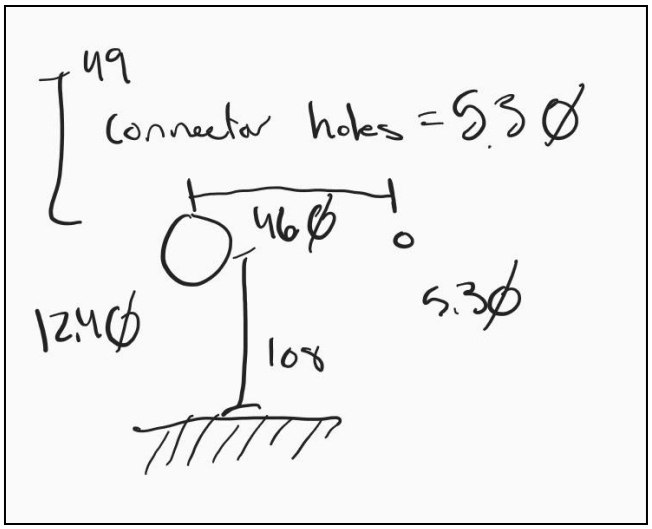




This was the trajectory we decided on after testing other link lengths and types of mechanism. It became the basis of our mid fidelity prototype. The link lengths were eventually altered slightly for the final prototype to accommodate the staircase better.

4.3 DESIGNING THE FRAME/ ITERATIONS





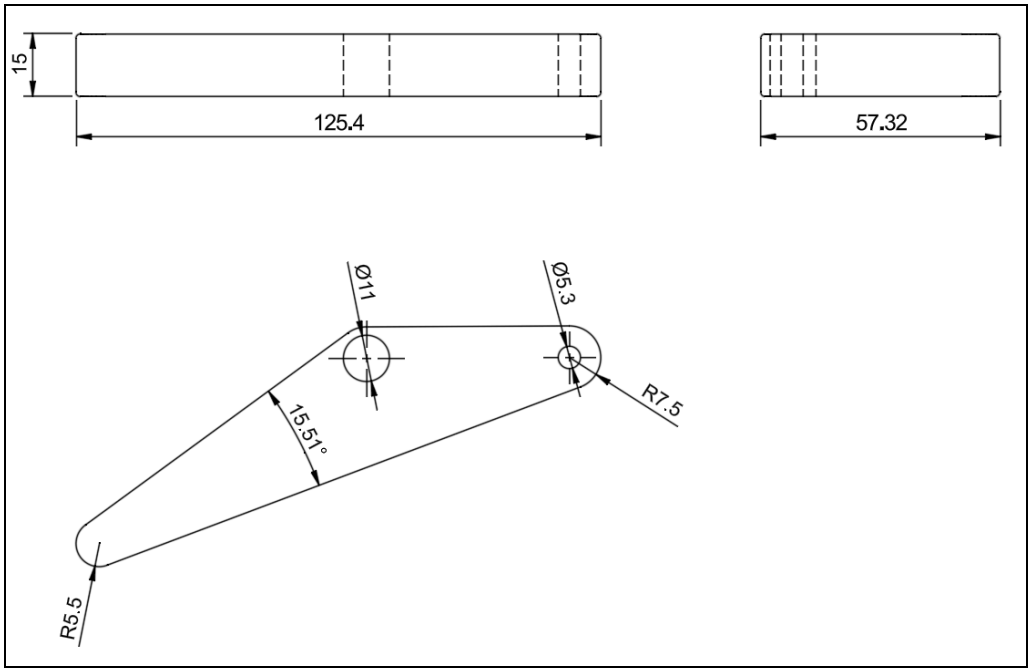
Initially we decided to split our frame into 3 parts so we could assemble the mechanism easily, but then realised the frame we designed was not wide enough to fit around the stairs, and changed our design to include a wider base. We also originally had two supports, but decided it was unnecessary.



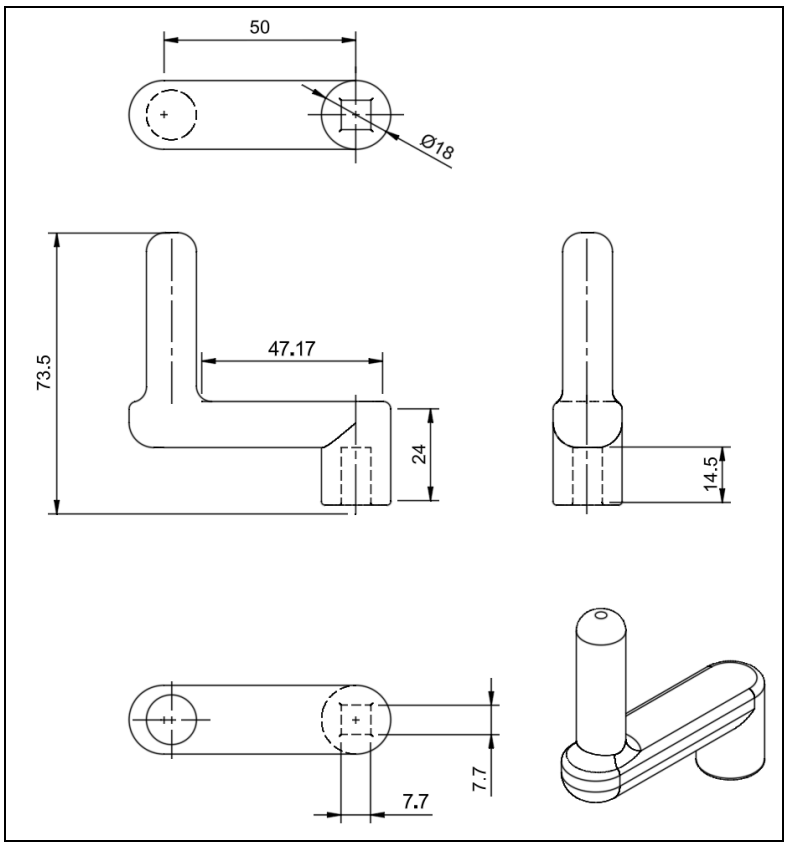


4.4 ENGINEERING DRAWINGS OF KEY PARTS

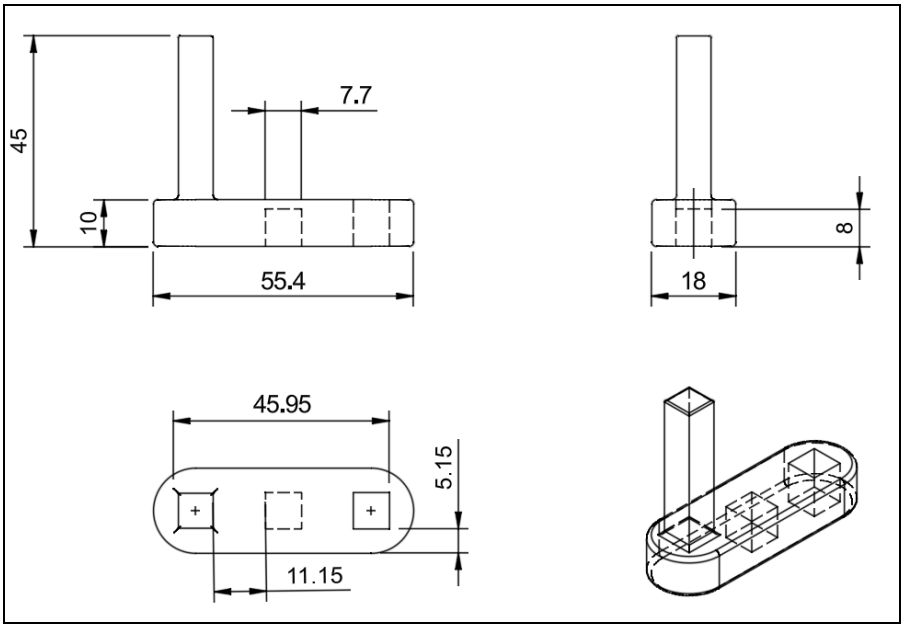
Drawing for Legs :



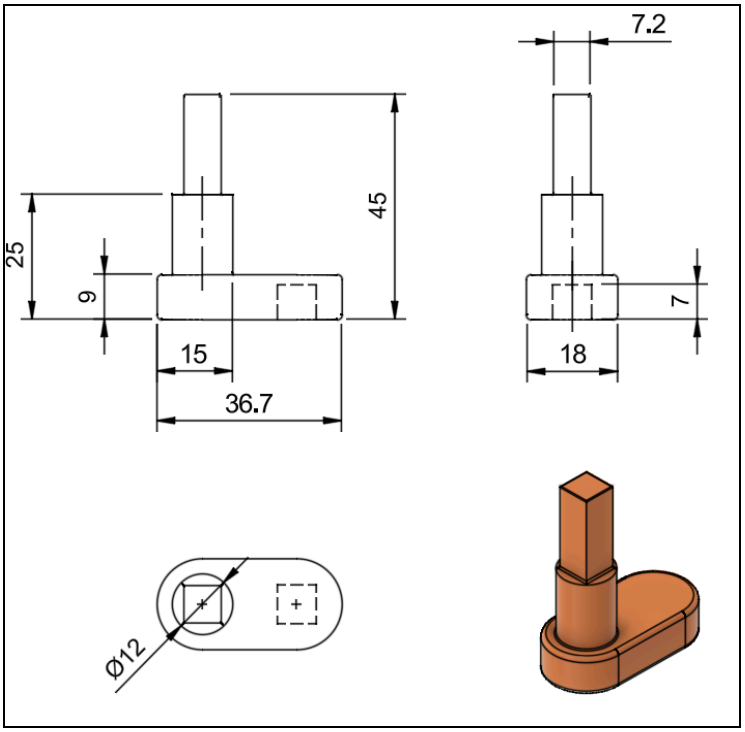
Drawing for Crank :



Drawing for Torque Transmitting Between Legs (Yellow Link) :



Drawing for Link Connected to Crank (Orange Link) :



4.5 ADDITIONAL SKETCHES

